Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Integrated Vehicle Control Unit (VCU) with Predictive Energy Management and Battery Stress Constraints to Maximise Regenerative Braking			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	Technology Field 1 (Large Category)		Technology Field 2 (Large Category)	
	Electrical & Electronics		Information & Communications	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Application and Utilization Technologies (HV, etc.) (Code 200903)	Software	Embedded SW (Code 300801)
Concept of Proposed Technology	Proposed technology integrates a production-intent Vehicle Control Unit (VCU) with model-based predictive supervisory control to maximise regenerative braking energy capture while enforcing battery stress, thermal, and safety constraints.			

	Core technical elements provided by TNO: • Modular Energy Management Strategy (MEMS): a modular, optimisation-based energy management suite that supports rapid evaluation of multiple powertrain configurations and can execute efficiently on vehicle control units. [1] • Battery state and ageing-aware constraints: State-of-X (SoX) estimation, Remaining Useful Life (RUL) concepts, and degradation-aware control, aligned with TNO battery management research and demonstrated in public projects (e.g., SeBaPaD secure battery passport demonstrator). [4][7] • Model-Based Systems Engineering (MBSE) and verification approach: co-design and systematic Verification and Validation (V&V) methodology as described in Steven Wilkins co-authored work, including guidance on model confidence for V&V. [3] • End-to-end test and validation infrastructure: climate battery pack and vehicle testing at 750 V / 400 A / 300 kW with temperature capability -60°C to +150°C and RH 5% to 95%, plus climatic altitude testing down to 628 mbar (4000 m). [5] The Korean SME partner contributes a VCU platform with integrated brake control intended to maximise regenerative energy and vehicle operability.
Rationale for Joint R&D Project	Why a joint R&D project under the 2026 TIPA International Joint R&D Program: • The Korean SME partner's VCU focus on integrated brake and regenerative control provides a direct application for TNO's predictive energy management and battery-aware constraint handling. • TNO provides validated modelling, BMS algorithm development toolchains (MIL and HIL), and large-scale environmental testing to de-risk performance and safety claims before productisation. [4][6] • Joint development enables a transferable reference architecture for VCU software that is reusable across xEV platforms and duty cycles, leveraging MEMS modularity and co-design methodology. [1][3] • The validation plan leverages TNO's Battery Lab and vehicle test infrastructure to benchmark behaviour under extreme temperatures, humidity, and altitude conditions. [4][5]
Key Words	1. Vehicle Control Unit (VCU) 2. Regenerative braking optimization 3. Modular Energy Management Strategy (MEMS) 4. Battery SoX and degradation-aware constraints 5. MIL/HIL and climate/altitude validation

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	1) Define a VCU-centred supervisory control architecture that coordinates traction, blended braking, and battery constraints with traceable requirements. 2) Implement predictive energy management using MEMS-derived optimisation and modular component libraries suitable for embedded execution. [1] 3) Integrate battery state estimation and degradation-aware constraints (SoX, RUL-related indicators) to protect lifetime and total cost of ownership. [4][6][7] 4) Establish a staged Verification and Validation (V&V) process from Model-in-the-Loop (MIL) and Hardware-in-the-Loop (HIL) to climate and altitude chamber validation. [5][6] 5) Deliver a demonstration on representative vehicle hardware showing stable operation under relevant environmental and electrical conditions.			
R&D Target Specifications	Target specifications are defined as verifiable test and deliverable criteria, with quantitative validation conditions based on TNO facility capability: • Validate VCU supervisory functions in climate pack/vehicle testing up to 750 V / 400 A / 300 kW, temperature -60°C to +150°C, RH 5% to 95%. [5] • Validate functional robustness in climatic altitude vehicle testing down to 628 mbar (equivalent to 4000 m) with wind generator capability (120 km/h @ 1 m²). [5][8] • Provide an MIL and HIL test suite for regression and calibration of energy management and braking-regen coordination algorithms, consistent with TNO's BMS algorithm development toolchain. [6] • Deliver a documented V&V dossier: requirements traceability, model confidence rationale, test procedures, and test results suitable for industrial transfer. [3]			
R&D Plan	R&D plan (example 36 months, can be adapted to TIPA call requirements): **Phase 1 (Months 0-12): Requirements and baseline models** • System requirements and interfaces (VCU, BMS, brakes, drivetrain). • Baseline vehicle and battery models; initial MEMS setup; MIL validation plan. Milestone: MIL baseline controller and test matrix agreed. **Phase 2 (Months 13-24): Algorithm development and HIL integration** • MEMS-based predictive energy management and blended braking optimisation. • Battery-aware constraint integration (SoX, degradation proxies) and safety monitoring. • HIL test bench integration and automated regression. Milestone: HIL-validated controller with documented behaviour under representative cycles.			

	Phase 3 (Months 25-36): Environmental validation and demonstrator package • Climate chamber and altitude chamber validation campaigns using TNO facilities. [5] • Robustness assessment, engineering guidelines, and transfer package for partner productisation. Milestone: Demonstrator and final V&V dossier delivered.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**TNO (Research Institute) roles and responsibilities:** • System and control architecture design for VCU-centred supervisory control. • MEMS-based predictive energy management development, calibration and embedded-ready implementation guidance. [1] • Battery modelling and BMS algorithm development support (SOC/SOH/SOF, thermal estimation, ageing models), including MIL/HIL toolchains. [4][6] • Execute V&V in TNO Helmond facilities including climate pack/vehicle testing and altitude climate testing. [5] • Project management and reporting on the overseas institute side; official contact: Frank Kupper. [8]
Expected R&R of Korean SME Partner	**Korean SME partner roles and responsibilities:** • Provide the production-intent VCU hardware and embedded software environment. • Provide braking and powertrain interface specifications and support integration for blended braking and regenerative energy maximisation. • Provide representative vehicle-level signals, constraints, and calibration support to enable realistic MIL/HIL setups. • Support prototype deployment and joint testing campaigns, including providing test vehicles or equivalent integration platforms. • Support industrialisation planning, including manufacturing constraints, cost targets, and compliance needs.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technological outcomes • A validated VCU-centred predictive supervisory control stack integrating regenerative braking optimisation, energy management, and battery-aware constraints. [1][4][6] • A reusable model and controller library plus MIL/HIL test suite supporting faster integration across multiple xEV platforms. [1][6] • Verified environmental robustness based on climate and altitude chamber testing at TNO Helmond. [5][8] Commercialisation and broader impact • A transfer-ready reference architecture and V&V dossier enabling integration into the partner's product roadmap and reducing productisation risk. • Improved competitiveness of Korean xEV control components through internationally benchmarked validation and documented lifetime-aware control approaches. Scientific impact • Joint results aligned with peer-reviewed open-access work by TNO and collaborators on co-design, verification and validation, and battery-aware optimisation. [3]

6. References

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[8] TNO. "Sustainable mobility and logistics" (mentions Altitude Climate Chamber up to 4000 m). https://www.tno.nl/en/sustainable/mobility-logistics/ (accessed 2026-01-23).

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization